

Individual Report

FIT5225 2026 S1 — Assignment 2

Name: Diya Vishalkumar Patel

Student ID: 36782378

1. My Role in the Project

My main job in the group was building the frontend. I was in charge of the whole React single page application the part users actually see and click through. It was quite a lot of work but I am happy with how it turned out.

I started with the login and sign up pages since everything else depended on authentication working first. I connected them to AWS Cognito by making direct API calls using fetch, without any extra libraries. Getting the JWT token handling right took me a while, I had to decode the token payload to pull the user's first and last name out of it to show on screen. The sign up form asks for first name, last name, email and password which was a specific requirement in the assignment spec. I also made sure that if someone tries to go to any page without being logged in, they get redirected back to the sign in page straight away.

Once auth was done I built the rest of the pages. The upload page has drag and drop and shows you the detected species after tagging. The search page has four different ways to query files by tag counts, by species name, by thumbnail URL, and by uploading a file to find similar ones. I also built the tag management page where you can bulk add or remove tags from multiple files at once, a delete page, and an alerts page for signing up to species notifications. For each page I wrote mock API functions so the UI was fully working even before Kiran finished the backend, which meant we weren't blocked waiting for each other.

I also looked after the GitHub repo for most of the project. There was a problem early on where a large file got into the git history and blocked the push to GitHub I had to use git filter-branch to clean it out. After that I made sure everyone had collaborator access and that commits were going up from all three of us. I also did the final testing on the live deployed URL and found that the notifications page wasn't connected to the backend yet, so I told the team about it so it could be fixed in time.

2. Teamwork Reflection

Honestly I think for a project this complicated we did pretty well as a team. None of us had done a multi-cloud assignment before so there was a learning curve for everyone, but we figured it out.

Working with Kalana was good. Kalana was handling the AWS setup and Cognito configuration, and sent me the User Pool ID, App Client ID and region fairly early which helped me a lot. Without those details I couldn't connect the real Cognito auth to my frontend I would have been stuck on mocks for much longer. Kalana also wrote most of the architecture section for the team report which was thorough and saved the rest of us a lot of time.

Kiran had a rough time with his laptop during the project and there were a few days where things slowed down on the backend side. But to his credit he kept the group chat updated and asked for help testing when he needed it rather than just going silent. I think that made a big difference. Because I had built the UI with mock responses from the start, we could still demo everything visually even while the real APIs were still being finished on his end.

One thing I wish we had done differently is talk about the API format earlier. When I was building the search pages I wasn't sure exactly what data the backend was going to

return like what the field names would be, whether URLs would come back as strings or objects, that kind of thing. It meant I had to make a few assumptions and then fix things later when the real responses came through. If we had spent like 30 minutes at the start just writing down the expected request and response formats for each endpoint it would have saved some back and forth.

Overall though I am really glad we got it working. The app is live on GCP Cloud Run, the login works with real Cognito, all the pages are there, and it looks good. This was genuinely one of the harder assignments I have done and I feel like I learned a lot from it especially around cloud authentication, deploying to Cloud Run and just trying to coordinate a system that spans two completely different cloud providers.